

Team 03: Research Question and Analysis Plan

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Research question

Does the level of liberal democracy in a country predict the share of its social science and humanities output devoted to politically sensitive disciplines (political science, sociology, law, international relations, public administration)?

Rationale

Autocratic regimes have strong incentives to suppress research that could challenge regime legitimacy or inform political opposition. If these incentives shape the composition of SSH knowledge production, we should observe systematic differences in what fields are studied — not just how much is published. This question directly addresses the “contents and direction” dimension of the overarching research question by examining whether autocracy distorts the disciplinary portfolio of SSH research toward politically safer fields.

Theoretical mechanism

Autocratic rulers face a legitimacy dilemma: they benefit from a productive higher-education sector (for economic development and international prestige), but research in certain SSH disciplines — political science, sociology, law, international relations — can produce findings that challenge regime narratives, document repression, or provide intellectual resources for opposition movements. Regimes manage this tension through multiple channels: direct censorship and topic restrictions at universities, strategic allocation of research funding toward “safe” fields (economics, business, psychology), and the cultivation of self-censorship norms among researchers who anticipate career costs for working on sensitive topics. The net effect is a compositional shift: autocracies produce a lower share of their SSH output in politically sensitive fields and a higher share in fields that are either politically neutral or regime-compatible. The expected direction is positive — higher values of `v2x_libdem` (more democratic) should predict a higher share of output in politically sensitive disciplines.

Theory family

regime-field-distortion

Estimand

The average change in the share of a country’s SSH articles published in politically sensitive fields associated with a one-unit increase in the liberal democracy index (`v2x_libdem`), holding constant

country-level time-invariant characteristics and global time trends.

Unit of analysis

Country-year (aggregated from article-country rows).

Outcome variable

Constructed: `share_sensitive` — the proportion of a country-year’s SSH articles (denominator: `n_articles_country_year`) whose `subject_primary` falls in a predefined set of politically sensitive fields (Political Science, International Relations, Sociology, Law, Public Administration, Social Issues, Ethnic Studies, Women’s Studies).

Key independent variable

`v2x_libdem` (V-DEM Liberal Democracy Index, continuous 0-1).

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Method

OLS regression with two-way fixed effects (country + year). The outcome is a country-year-level proportion constructed by aggregating article-level field classifications. No text analysis is required — the outcome is derived from the structured `subject_primary` variable.

Model specification

Data construction

1. Define politically sensitive fields: Political Science, International Relations, Sociology, Law, Public Administration, Social Issues, Ethnic Studies, Women’s Studies.
2. For each article-country row, code `sensitive = 1` if `subject_primary` is in the sensitive set, 0 otherwise.
3. Collapse to country-year level: compute `share_sensitive` = mean of `sensitive` (i.e., number of sensitive-field articles / total SSH articles for that country-year).
4. Restrict to country-years with at least 20 SSH articles (to avoid noisy proportions from very small counts).
5. Restrict to 1990-2023 (pre-1990 WOS coverage is unreliable per PLAN.md).

Main specification

$$\text{share_sensitive}_{\{c,t\}} = \beta * \text{v2x_libdem}_{\{c,t\}} + \alpha_c + \gamma_t + \epsilon_{\{c,t\}}$$

- **Outcome:** `share_sensitive` (proportion, 0-1)
- **Key predictor:** `v2x_libdem` (continuous, 0-1)
- **Fixed effects:** Country (`alpha_c`) + Year (`gamma_t`)
- **Standard errors:** Clustered at the country level
- **Additional controls:** `log(n_articles_country_year)` — controls for the scale of a country’s research system, which may independently affect field composition

Expected coefficient

$\beta > 0$: more democratic country-years have a higher share of SSH output in politically sensitive fields.

Causal identification strategy

Variation exploited

Two-way fixed effects exploit within-country, over-time variation in `v2x_libdem`. The identifying variation comes from countries that experience changes in their level of democracy over the 1990-2023 period — democratizing or autocratizing episodes.

Confounders controlled

- **Country FE:** Absorb all time-invariant country characteristics (geography, colonial history, language, cultural traditions in SSH, size of higher education sector at baseline).
- **Year FE:** Absorb global trends (worldwide expansion of political science as a discipline, shifts in WOS indexing coverage, global funding trends).
- **Log total output:** Controls for the possibility that countries with more total SSH articles mechanically have different field distributions.

Identification threats

1. **Time-varying economic development:** GDP growth may correlate with both democratization and expansion of specific SSH fields. Not directly controlled (GDP is not in the corpus), but country FE absorb baseline development and year FE absorb global growth trends.
2. **Internationalization of higher education:** Countries that democratize may simultaneously integrate into global academic networks, which could independently shift field composition toward Western-norm disciplines. This is a plausible confounder that cannot be fully separated from the regime effect.
3. **WOS indexing bias:** WOS may start indexing more journals from a country as it democratizes and internationalizes, and newly indexed journals may disproportionately cover political science or sociology. This would bias estimates upward. We partially address this by controlling for total output volume and restricting to post-1990.
4. **Reverse causality:** In principle, a vibrant political science community could contribute to democratization. This is unlikely to drive results at the country-year level given the slow-moving nature of both academic field composition and regime change, but cannot be ruled out.

Robustness checks

1. **Alternative regime measure:** Replace `v2x_libdem` with `regime_binary` in the main specification.
2. **Alternative threshold for minimum articles:** Use 50 instead of 20.
3. **Narrower definition of sensitive fields:** Restrict to Political Science, Sociology, and Law only (dropping the smaller categories).

Expected output files

File	Description
figures/descriptive_shares_by_regime.png	Bar chart: mean share of sensitive fields by <code>v2x_regime</code> category
figures/main_regression_table.png	Regression table: main specification + robustness checks
figures/trend_share_sensitive.png	Line plot: share of sensitive fields over time, by regime type
figures/coefplot_robustness.png	Coefficient plot comparing estimates across specifications